

IOT Enhanced Robotic Waste Collector With Blynk App Control and Metal Proximity Sensing

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Abstract— Efficient metal waste collection and recycling are vital for reducing environmental risks in trash management. Metal waste constitutes a significant portion of overall waste, posing threats to ecosystems when improperly disposed of. Our proposed robotic waste management system utilizes an ESP32 microprocessor for efficient recognition and classification of metallic and non-metallic materials, thereby enhancing sorting accuracy. Metal proximity sensors enable meticulous separation, crucial for mitigating environmental risks associated with metal waste. Integration of the Blynk Android application facilitates real-time control and monitoring, engaging users in the process. This collaborative hardware-software ecosystem enables directed deposition of materials into designated bins, expediting waste disposal and facilitating subsequent recycling efforts. Our approach offers a creative solution to accelerate and improve metal waste sorting, contributing to environmentally friendly waste management practices and ensuring efficient material segregation for long-term recycling initiatives. By leveraging innovative technology, our system addresses the pressing need for expedited and enhanced metal waste management, offering a sustainable solution to minimize environmental impact while promoting responsible recycling practices.

Keywords – ESP32 Microprocessor, metal proximity sensor, Blynk Android application, metal and non-metal segregation

I. INTRODUCTION

Waste is an unavoidable consequence of human activity in the commercial, industrial, and domestic domains. The amount and variety of trash created has increased to an unprecedented degree as a result of the acceleration of industrial expansion and urbanization. Modern waste streams are so large and complicated that they are beyond the capacity of traditional waste management methods, which usually depend on dumping and burning. Our approach is based on a comprehensive view of waste handling, utilizing a self-contained robotic system intended for effective material separation.

A. Current Challenges in Waste Management

The primary challenge in today's garbage or metal trash collection is the traditional method of gathering waste at one spot and then delivering it to a specific unit for trash segregation or separation. The two stages of this conventional procedure are gathering all trash kinds in one location and then sorting the garbage according to whether it is metallic or non-metallic. This methodical approach has some shortcomings and inefficiencies. First off, there is an environmental impact and an increase in operational costs due to the additional resources required for the transportation of mixed trash to a segregation unit. Furthermore, there is a delay before the actual sorting process starts due to the time it takes to transport garbage to a different facility for segregation.

B. Environmental Impact of Uncollected Waste

Uncollected metal waste poses a serious threat to the environment that has far-reaching effects. By the year 2025, the world's annual production of metal waste is expected to reach 2 billion metric tons, which will contaminate the land and water. Metals like lead and mercury leak into the environment due to improper disposal and a lack of an effective waste management infrastructure. This puts human health in danger and puts ecosystems at risk of long-term harm and diseases. Furthermore, according to estimates from the World Health Organization, environmental hazards—including those related to unregulated metal waste—cause more than 12 million deaths yearly. Immediate and creative action is needed to improve garbage collection, segregation, and recycling procedures worldwide to address this problem.

II. ROLE OF IOT IN ENHANCING WASTE MANAGEMENT

Waste management is going through a new era as a result of the integration of the Internet of Things (IoT).

A. Automation in Waste Segregation

Waste segregation procedures are greatly improved by the automation that comes with integrating IoT into waste management systems [1]. Waste collection and sorting have grown more efficient and intelligent with the use of smart sensors and networked devices.

Smart Sensors for Material Identification: Metal proximity sensors [2] and other IoT-enabled smart sensors help automate the trash sorting process. These sensors can differentiate between metallic and non-metallic components [3] [4] when determining the material of waste products. The incorporation of these sensors into the robotic arms of our trash collector enables real-time junk detection.

Efficient On-Site Segregation: The Internet of Things-enabled automated trash segregation method reduces the need for post-collection segregation [5]. The waste collector can reduce the time and resources needed for sorting by performing on-site segregation by smart sensors. This ultimately results in a waste management workflow that is quicker and more effective to transport garbage to a different facility for segregation.

B. Integration of IoT and Blynk for Smart Control

The interplay between the Blynk platform [6] [7] and the Internet of Things improves the intelligent control capabilities of waste management systems.

ESP32 Microcontroller for Connectivity: With integrated Bluetooth and Wi-Fi modules, the ESP32 microcontroller [8] is essential for the garbage collector to establish connectivity. With the help of this microcontroller, the garbage collector may connect to the Internet of Things ecosystem and exchange and regulate data in real time.

Blynk Application for User Interaction: The flexible IoT platform Blynk is used to create an intuitive waste collection interface. Users may remotely operate and keep an eye on the garbage collector in both manual and automatic modes using the Blynk application. User involvement is improved by the user-friendly interface, which makes it easy for people to control the robot, keep an eye on bin levels, and get alerts [9].

III. METHODOLOGY

The primary goal of this robot is to take trash collected at the workplace and parks to determine whether they are metal or not, and then place them in appropriate bins so that the collected metal trash can eventually be sent for recycling.

A. System Architecture

Figure 2 shows the block diagram of the robotic waste collector which consists of motors, sensors, a micro-controller unit, and a metal proximity sensor. In the proposed system, all the components are controlled by the ESP32 microcontroller which has a dual-mode Bluetooth and integrated Wi-Fi. The power supply for the entire robot is given by a 12v Lithium-Ion battery. A bin is attached to the robotic model which is divided into two sections, one for metal trash and another for non-metal trash.

The movement of the robot is achieved by 4 BO motors attached to the wheels. The robot has 2 ultrasonic sensors in front of the chassis for obstacle detection [10] [11]. The robot has two robotic manipulators. Two servo motors in the manipulator control the sideways movement of the arm to grab the trash. The grabbing part of the arm has a metal inductive sensor to determine whether the grabbed scrap is metal or non-metal. The rotation of the bin is controlled by a servo motor. The servo motor [12] can rotate 180° based on the detection by the Metal proximity sensor. Two other servo motors control the lifting of the arm up and down to place the trash in the appropriate section of the bin. Each section of the bin has ultrasonic sensors to monitor the level of the bin continuously. The levels are shown in the Blynk app for the user. The overall structure of the robot is shown in the figure 1.

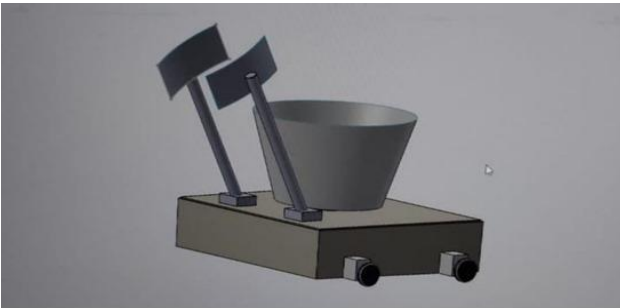


Fig. 1. Structure of Robotic Waste Collector [5]

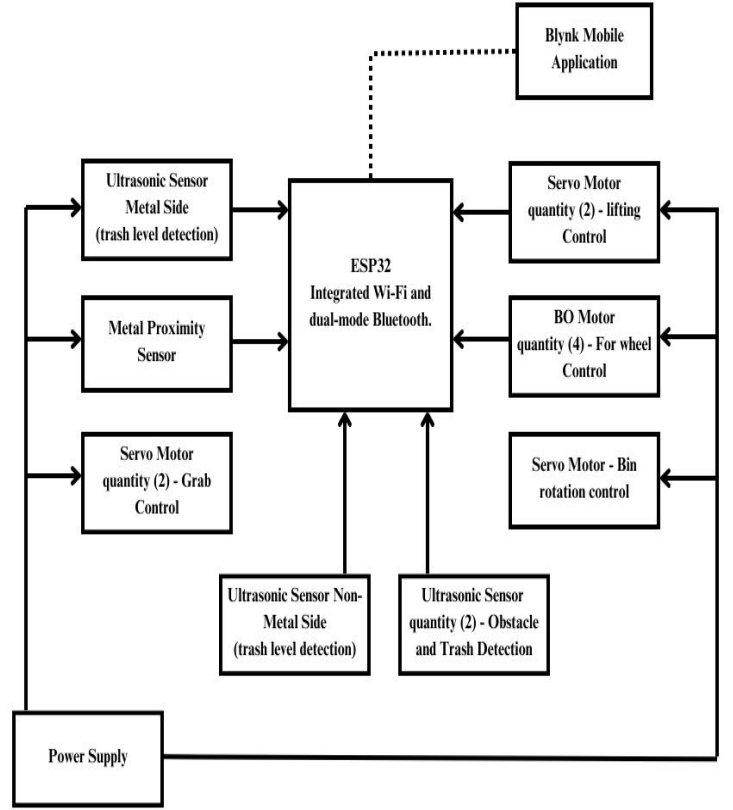


Fig. 2. The Block Diagram

B. Overview of Robotic System Components

The hardware components used are: ESP32-WROOM-32: The ESP32 serves as the central processing unit, managing various components and facilitating wireless communication with the Blynk app. Its dual-core architecture enhances multitasking capabilities, while integrated Wi-Fi and Bluetooth support enable real-time control and data transfer.

BO Motor: This provides the necessary mechanical power for the movement of the robot by converting electrical energy into mechanical energy. BO motors propel the robot, allowing it to move forward, backward, turn, and navigate in response to user commands or autonomously avoid obstacles.

Motor Driver: The motor driver regulates voltage and current to control the speed and direction of BO motors in the robotic system. Managing current flow, it prevents motor damage, and they are used for motor interfacing.

Servo Motor: Servo motors control various components, including arm movements, bin rotation [15], and arm elevation. A servo motor can rotate with great precision.

Ultrasonic sensor: Ultrasonic sensors are primarily used as level sensors [13] [14]. In our project, the ultrasonic sensor is used to detect the level of waste filled in the trash bin. They are also used for obstacle and trash detection. It is a transducer as it can convert the electrical signal into sound waves and vice versa.

Metal proximity sensor: Metal proximity sensor (inductive proximity sensor) can detect metal targets near the sensor without any physical contact with the metal object [16]. This is used to detect if the trash picked is metal or non-metal.

Power Supply: The 12V Lithium-Ion battery powers the entire robot, providing a reliable and portable energy source for sustained operation.

The software tools used are:

Arduino IDE: The Arduino IDE is used to write computer code in C or C++ and upload this code to the physical board. This software is used to program the robotic waste collector to autonomously identify, collect, segregate, and navigate the environment.

Blynk App: Blynk [17] enables us to create interfaces for controlling and monitoring hardware. The Blynk platform has three main elements:

- a) **Blynk app builder:** It enables us to use different widgets to develop the project.
- b) **Blynk Server:** This oversees facilitating the connection between the hardware and our mobile device.
- c) **Blynk Libraries:** Facilitates communication with the server.

C. Working Modes

i. Automatic Mode

In the automatic mode of our robotic waste collector, we implement obstacle detection and avoidance through a dual ultrasonic sensor setup. Two ultrasonic sensors, labeled A and B, are strategically arranged in two rows at the front section of the robot. As the robot traverses its path, the ultrasonic sensors continuously scan the environment, aiming to discern between obstacles or objects and potential trash.

Ultrasonic sensor A is positioned in the lower part of the chassis, while ultrasonic sensor B is placed a few centimeters above sensor A as shown in figure 3. When both ultrasonic sensors detect an object or obstacle, the robot's response varies based on the nature of the obstruction. If the sensors (A and B) collectively identify a solid object or a wall, the robot initiates a turning mechanism to navigate around the obstacle, giving preference to sensor B. It then turns right and checks for obstacles. If an obstacle is detected, the robot returns to the center point and takes a left turn to explore further for trash. Likewise, if the obstacle is detected solely by ultrasonic B, the robot promptly turns and takes another path, executing the right-turn-and-check-center-left-turn process.

This design choice allows the system to give precedence to readings from sensor B, considering them more useful in identifying potential obstacles. Additionally, if an object is detected only by sensor A, it is trash, with the exception that in outdoor environments such as parks and gardens, where lengthy grasses are prevalent, the system may inaccurately identify them as trash. This approach ensures that the robotic waste collector effectively navigates through its environment, distinguishing between obstacles that warrant avoidance and trash that merits collection.

ii. Manual Mode

In the manual mode, instead of relying on autonomous decision-making, the robot follows user controls transmitted through the Blynk app interface. Obstacle detection and avoidance are integrated using a mobile app using user commands. For real-time communication between the app and the robot, the ESP32 maintains a continuous connection with the Blynk server. The robotic waste collector utilizes a Wi-Fi module for communication which offers advantages over Bluetooth due to its longer range and enhanced connectivity stability. Wi-Fi operates on a higher frequency, enabling faster transmission of sensor data and user commands, crucial for real-time responsiveness.

The app empowers users to command the robot's movement—forward, backward, left, and right—as well as control a servo-operated arm for efficient trash collection. While the ultrasonic sensors are still active, their data does control the robot's autonomous actions in manual mode. Instead, the user's commands take precedence, enabling an interactive and user-driven experience. The user interface on the Blynk app is designed to include buttons, sliders, or any other widgets that will help the user control the robot. User inputs received from the Blynk app are mapped to specific actions to be performed by the robotic waste collector. For example, if the user presses a button to move the robot forward, corresponding with the use of code this command is interpreted by the ESP32 and controls the motors for forward movement accordingly.

D. Steps Involved

i. Trash Collection and Segregation

Before delving into the trash identification and collection process, it's crucial to note that whether the trash is collected autonomously or manually, the subsequent step involves determining if the trash is of metal or non-metal composition. robot picks up the trash from the ground using its modified dustbin manipulator (i.e., a dustbin cut into two halves). Each end of the cut dustbin is attached to the corresponding end of the manipulator.

Trash Identification: Once the robot utilizes its arms to pick up the trash, the following step involves trash identification as shown in figure 3. Our project incorporates a metal proximity sensor within the robot's arm, i.e., inside the dustbin manipulator. The metal proximity sensor assists in identifying whether the trash is metallic or non-metallic.

Trash Collection: A bin is divided into two sections for metal and non-metal items, is attached to the robot. After the trash is identified using the metal proximity sensor, if the waste is metallic, it is placed inside the bin. However, if the waste is non-metallic, the bin is rotated 180° to collect the non-metallic waste.

ii. Bin Level Monitoring Using Blynk App

The Bin Control and Level Monitoring [18] aspect of our waste management system is integral to ensuring efficient and timely waste disposal. The bin is divided into two sections, segregating collected scraps based on their material composition:

Section 1: For metal trash (one-half of the bin)

Section 2: For non-metal trash (the other half of the bin)

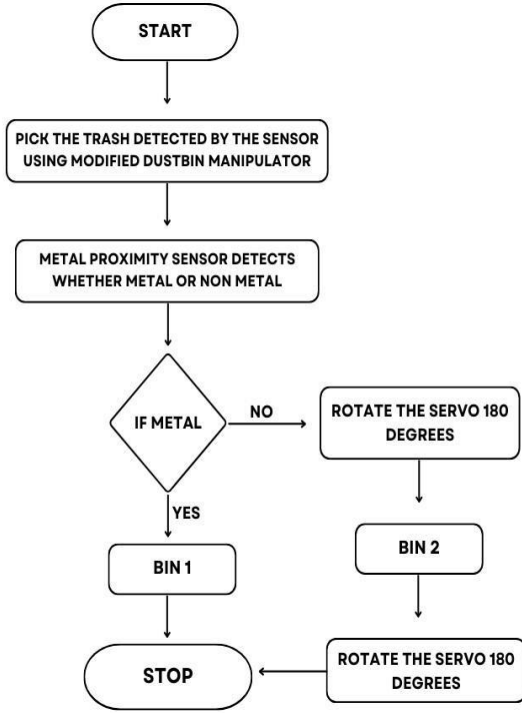


Fig. 3. Flowchart for Trash Identification

Continuous monitoring of the filled levels within both sections is achieved through the implementation of ultrasonic sensors. To ensure precise and accurate level monitoring, sensors are placed on the lid of the bin, each one on both segregated sections. These sensors accurately gauge the real-time levels of metal and non-metal trash present in the bin. Predetermined thresholds are set for each section of the bin. These thresholds represent the optimal fill capacity, ensuring the bin can handle waste effectively. The system takes into account the capacity constraints of the bin and ensures that it doesn't exceed its designated limits.

Once the predetermined threshold is crossed in either section, the system triggers notifications, promptly alerting the user to take necessary action. This timely notification serves as a proactive reminder to empty the bin. The implementation of this smart level monitoring system not only optimizes the usage of the bin but also offers users a hassle-free and efficient approach to waste management. By providing real-time updates and proactive alerts, the system promotes a user-centric and environmentally conscious waste disposal process.

IV. RESULTS AND DISCUSSION

A. Simulation Results

Figure 4 illustrates the simulation results of the DC motors (responsible for the robot's wheel movement) and servo motors utilized for various functions such as rotating the dustbin and handling the pick-up and carry of waste. There are four BO motors shown in Fig 4, each depicting the wheels placed on the chassis of the robot, enabling movement in different directions: forward ('F'), backward ('B'), left ('L'), right ('R'), and stop ('S').

Servo motors play a key role in controlling different parts of the robotic arm and the rotation of the dustbin. Servo motors A, in Figure 4, pick up trash from the ground. They control the ends of the modified dustbin manipulator during the trash-picking process. Servo motors B, highlighted in Figure 4, help carry waste from the ground to the bin. Servo motor C, shown in Figure 4, rotates the dustbin for efficient metal and non-metal segregation.

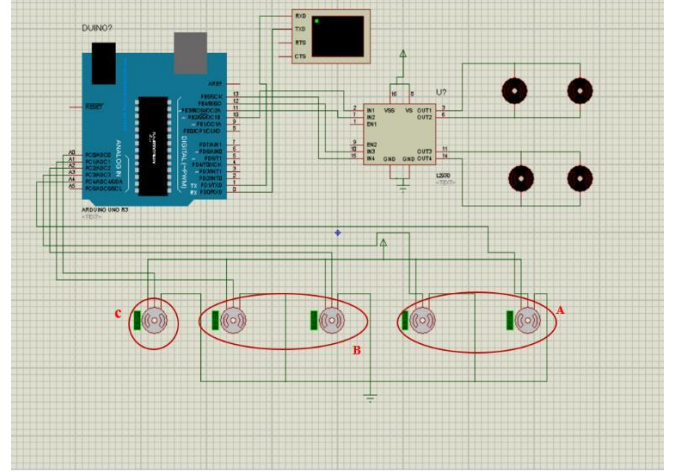


Fig. 4. Simulation of BO and Servo Motors

B. Sensor Calibration

The robotic waste management system's actuators and sensors were calibrated to guarantee accurate and dependable operation.

Ultrasonic Sensor Calibration: The ultrasonic sensors were calibrated to measure distances for obstacle detection and bin level monitoring precisely. This involved calculating the speed of sound in the current environmental conditions. The formula used for distance measurement was:

$$d = \frac{s \times t}{2}$$

Where,

d = Distance to an object

s = Speed of sound

t = Time taken by the ultrasonic signal to travel to the object and back

Once calibrated, the sensors could detect objects within a specified threshold limit. The ultrasonic sensor's Tx emitted a signal, which, upon striking an object, reflected to the Rx. The gap between the sensor and the object was determined as the distance, and when it reached the predefined threshold, the system initiated further actions.

Precise and controlled movements of the robotic arm and bin rotation were achieved by calibrating servo motors. The process of calibration included figuring out which rotational angle corresponded to specific control signals. Angle calibration does map control signals to specific angles of rotation. Verified the accuracy of the servo motors in achieving the desired angular positions.

C. Robot Chassis Setup

The robot's chassis setup is shown in Figure 5 and Figure 6. The robot's chassis is made of plywood which is around 25.3 cm long [19] and has four wheels at the bottom. BO motors are attached to each of the four wheels. Two wooden boards are designed with a center gap to securely hold the ESP32, Motor Driver, and connecting wires. A hole in the middle of the lower board is created to allow the wires to pass through neatly to power the BO Motors. On the top board, there's a servo motor that is placed with a circular wooden platform on which a bin can be placed. This setup allows the servo motor to rotate the bin, facilitating the separation of metal and non-metal items. There are two ultrasonic sensors present in front of the chassis to facilitate obstacle detection and avoidance.



Fig. 5. Robot Chassis setup – Front view



Fig. 6. Robot Chassis setup – Side view

D. Blynk App Interface

In Figure 7 of the Blynk app interface, the "obstacle" widget is highlighted, representing ultrasonic sensor B for obstacle detection. The "wastage" widget corresponds to ultrasonic sensor A, which detects trash. The "Forward" and "Backward" widgets control wheel movement. The "collect wastage" widget facilitates picking up waste using the modified dustbin manipulator from Section 2.3.1, enhancing waste management efficiency [20] and promoting a cleaner environment.

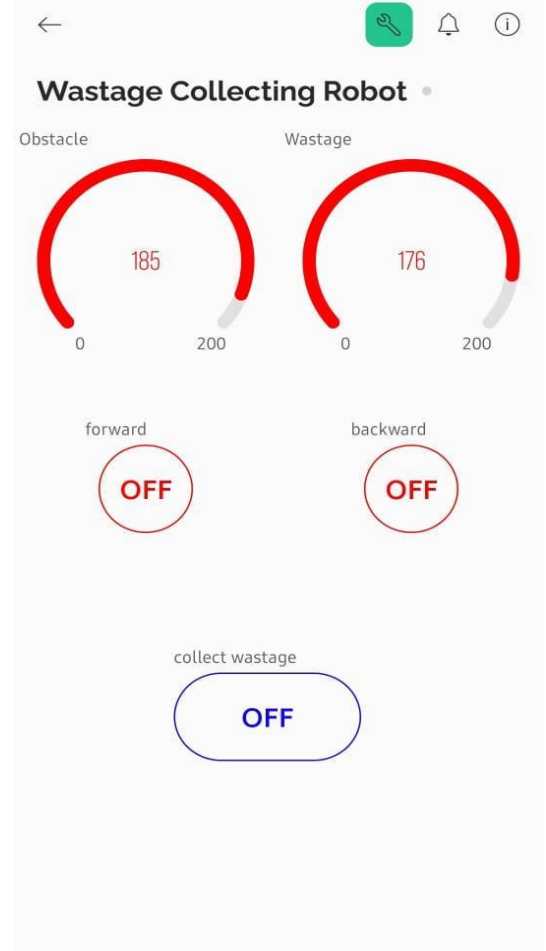


Fig. 7. Blynk Application Interface

V. CONCLUSION

Addressing the challenges of metal waste management is crucial for environmental sustainability. The segregation of metal and non-metal waste on-site using inductive proximity sensors further enhances waste management efficiency by facilitating automated sorting processes. Overall, leveraging IoT technology for waste management offers substantial benefits such as improved waste collection efficiency and enhanced real-time segregation through real-time monitoring and optimization of bin waste levels.

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